

DATA ANALYSIS INTERNSHIP

DAY 11 REPORT

Home Task – Graph Plotting using Matplotlib

Introduction

Matplotlib is a **Python visualization library** used for generating graphs and charts from numerical data. It allows data to be represented visually, making analysis and interpretation easier. Matplotlib is commonly applied in **Data Analytics, Machine Learning, Artificial Intelligence, and Data Science projects**.

Import Statement

```
import matplotlib.pyplot as plt
```

Explanation

- **matplotlib** → Main Python library used for visualization
 - **pyplot** → Module responsible for plotting graphs
 - **plt** → Shortcut name used for simpler coding
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Home Task

Creating Random Variables and Plotting Different Graphs

Various graph plotting methods were practiced using **Matplotlib in Google Colab** to understand graphical representation of data.

1. Line Graph

Program

```
import matplotlib.pyplot as plt
```

```
x = [1,2,3,4,5]
y = [10,20,30,40,50]
```

```
plt.plot(x,y)
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.title("Line Graph")
plt.show()
```

Purpose

A line graph is used to represent trends or continuous changes in data.

Functions Used

- `plt.plot()` → Draws the line graph
 - `plt.xlabel()` → Sets x-axis label
 - `plt.ylabel()` → Sets y-axis label
 - `plt.title()` → Adds graph heading
 - `plt.show()` → Displays the graph
-

2. Bar Graph

Program

```
import matplotlib.pyplot as plt
```

```
x = ["A","B","C","D"]
y = [20,40,60,80]
```

```
plt.bar(x,y)
plt.xlabel("Categories")
plt.ylabel("Values")
plt.title("Bar Graph")
plt.show()
```

Purpose

Bar graphs are mainly used for comparing values between categories.

Functions Used

- `plt.bar()` → Creates vertical bars

- `plt.xlabel()` → Adds x-axis label
 - `plt.ylabel()` → Adds y-axis label
 - `plt.title()` → Sets graph title
 - `plt.show()` → Displays output
-

3. Scatter Plot

Program

```
import matplotlib.pyplot as plt
```

```
x = [10,20,30,40,50]  
y = [15,25,35,45,55]
```

```
plt.scatter(x,y)  
plt.xlabel("X Values")  
plt.ylabel("Y Values")  
plt.title("Scatter Plot")  
plt.show()
```

Purpose

Scatter plots are useful for identifying relationships between two variables.

Functions Used

- `plt.scatter()` → Creates scatter plot
 - `plt.xlabel()` → Labels x-axis
 - `plt.ylabel()` → Labels y-axis
 - `plt.title()` → Adds graph title
 - `plt.show()` → Displays graph
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4. Pie Chart

Program

```
import matplotlib.pyplot as plt
```

```
values = [20,30,25,25]  
labels = ["A","B","C","D"]
```

```
plt.pie(values, labels=labels)
plt.title("Pie Chart")
plt.show()
```

Purpose

Pie charts are used to show percentage distribution of values in a circular format.

Functions Used

- `plt.pie()` → Creates pie chart
 - `plt.title()` → Adds title to chart
 - `plt.show()` → Displays chart
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Conclusion

On Day 11, different **graph plotting techniques** were practiced using **Matplotlib in Google Colab**. Multiple visualization methods including **line graphs, bar charts, scatter plots, and pie charts** were created to understand data representation more effectively. This session helped improve knowledge in **data visualization, graphical comparison, and plotting techniques using Python**.